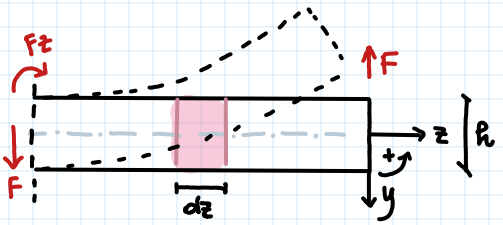
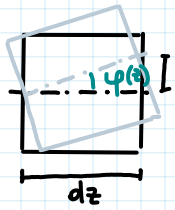
EQ^{ne} LINEA ELASTICA → TRAVE alla EULERO BERNOULLI

Quando abbiamo parlato di M_x, abbiamo introdotto una grandezza caratteristica: χ CURVATURA ELASTICA

$\chi = \frac{M_x}{EI_x}$ in generale $\Rightarrow$ $\chi = \frac{M}{EI}$ e avevamo anche dimostrato che $\chi = \frac{d\varphi_z}{dz}$

Considero adesso un elemento trave inflesso.





dv_M generato a seguito di M, mentre assumo un eventuale dv_T trascurabile (trascuro l'effetto di T) → approssimazione adeguata se la trave è snella ($l \gg h$)

Se considero solo il contributo di M:

$dv_M = -\varphi(z)\,dz$ (v è ⊕ verso il basso, concorde con y)

↓

posso anche scrivere $dy = -\varphi(z)\,dz$ $\quad \frac{dy}{dz} = -\varphi(z)$ $\quad$ Poi mi ricordo che $d\varphi(z) = \chi\,dz$

$$\begin{cases} \frac{d^2y}{dz^2} = -\chi = -\frac{M}{EI} \\ \frac{d\varphi}{dz} = \chi = \frac{M}{EI} \end{cases}$$

$\frac{d^2y}{dz^2} = -\frac{M}{EI}$ Eq^{ne} linea elastica del 2° ordine per una trave alla Eulero-Bernoulli

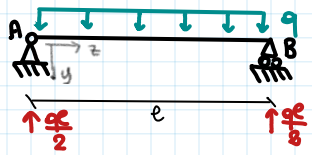
$$\left(\begin{aligned} y'' &= -\frac{M}{EI} \\ y''' &= -\frac{T}{EI} \\ y^{IV} &= \frac{q(z)}{EI} \end{aligned} \right)$$

NB
$y'' = -\chi$
$y' = -\varphi$
y = ⊕ se va verso il basso.

Per risolvere la linea elastica di una trave inflessa, e quindi ricavare la DEFORMATA (abbassamenti e rotazioni di tutti i pti) devo conoscere le condizioni a contorno → condizioni sulla CINEMATICA dei VINCOLI

Esempi · calcolare abbassamenti/rotazioni e tracciare deformata qualitativa

1



Devo conoscere $M(z)$

$M(z) = \frac{ql}{2}z - qz^2/2$ $\qquad y'' = -\frac{M}{EI} = \frac{1}{EI}\left(-\frac{qlz}{2} + \frac{qz^2}{2}\right)$

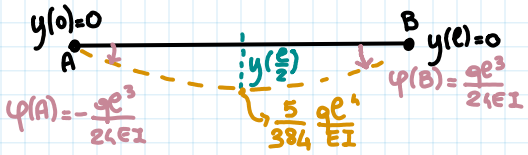
$y' = \frac{1}{EI}\left(-\frac{qlz^2}{4} + \frac{qz^3}{6}\right) + A$

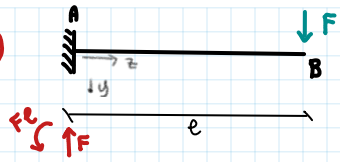
$y = \frac{1}{EI}\left(-\frac{qlz^3}{12} + \frac{qz^4}{24}\right) + Az + B$

Condizioni a contorno:
$\begin{cases} y(z=0) = 0 \\ y(z=l) = 0 \end{cases}$ $\quad B = 0$

$y(z=l) = \frac{1}{EI}\left(-\frac{ql^4}{12} + \frac{ql^4}{24}\right) + Al = 0$ $\qquad A = +\frac{ql^3}{24EI}$

- abbassamento in $\frac{l}{2}$: $y = \frac{1}{EI}\left(-\frac{ql}{12}\frac{l^3}{8} + \frac{q}{24}\frac{l^4}{16}\right) + \frac{ql^3}{24EI}\cdot\frac{l}{2} = \frac{1}{EI}\left(-\frac{ql^4}{96} + \frac{ql^4}{384} + \frac{ql^4}{48}\right) = \frac{1}{EI}\left(\frac{-4+1+8}{384}ql^4\right) = \frac{5}{384}\frac{ql^4}{EI}$ ↓⊕
- rotaz in A (z=0) ⊕, $\varphi = -y' = -A = -\frac{ql^3}{24EI}$ ⊕↻
- rotaz in B (z=0) ⊕↺ $\varphi = -y' = \frac{1}{EI}\left(+\frac{ql^3}{4} - \frac{ql^3}{6}\right) - \frac{ql^3}{24EI} = 0$ $\quad \frac{ql^3}{12EI} - \frac{ql^3}{24EI} =$ ⊕↺ $\frac{ql^3}{24EI}$



② 

$$M(z) = -F\ell + Fz$$

$$y''(z) = \frac{1}{EI}\left(-M(z)\right) = \frac{1}{EI}\left(F\ell - Fz\right)$$

$$y'(z) = \frac{1}{EI}\left(F\ell z - F\frac{z^2}{2}\right) + A$$

$$y(z) = \frac{1}{EI}\left(F\ell\frac{z^2}{2} - F\frac{z^3}{6}\right) + Az + B$$

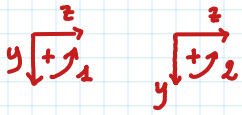
Condizioni a contorno:

$$\begin{cases} y(z=0) = 0 & \rightarrow B=0 \\ y'(z=0) = 0 & \rightarrow A=0 \end{cases}$$

$$\varphi(B) = -y'(z=\ell) = -\frac{1}{EI}\left(F\ell^2 - \frac{1}{2}F\ell^2\right) = -\frac{1}{2}\frac{F\ell^2}{EI} \circlearrowright$$

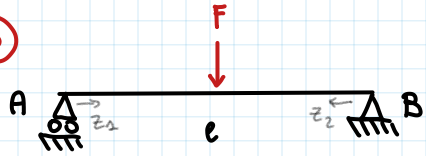
$$\eta(B) = y(z=\ell) = \frac{1}{EI}\left(\frac{F\ell^3}{2} - \frac{F\ell^3}{6}\right) = \frac{1}{EI}\left(\frac{2}{6}F\ell^3\right) = \frac{F\ell^3}{3EI} \downarrow$$

! ATTENZIONE: i segni quando scrivo l'eq^{ne} della linea elastica:



$\varphi_1 = \varphi_2$ stesso verso +

$\varphi_1 = -\varphi_2$ verso opposto

③ 

NB Devo spezzare la z e studiare 2 sistemi.
→ perché avrò 2 M(z) ≠

4 costanti di integrazione

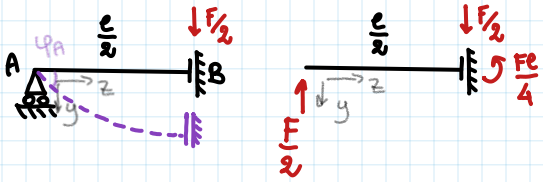
Condizioni a contorno:

$$\begin{cases} y(z_1=0) = 0 \\ y(z_2=0) = 0 \\ y(z_1=\frac{\ell}{2}) = y(z_2=\frac{\ell}{2}) \\ y'(z_1=\frac{\ell}{2}) = -y'(z_2=\frac{\ell}{2}) \end{cases}$$

OPPURE

Sfrutto la simmetria:

$$\begin{cases} y(z=0) = 0 \\ y'(z=\frac{\ell}{2}) = 0 \end{cases}$$



$$M(z) = \frac{F}{2}z$$

$$y'' = -\frac{F}{2}z \cdot \frac{1}{EI}$$

$$y' = -\frac{F}{2}\frac{z^2}{2} \cdot \frac{1}{EI} + C \qquad y'(z=\frac{\ell}{2}) = -\frac{F}{4}\frac{\ell^2}{4}\frac{1}{EI} + C = 0 \qquad C = \frac{F\ell^2}{16EI}$$

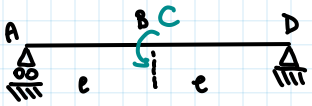
$$y = -\frac{F}{4}\frac{z^3}{3} \cdot \frac{1}{EI} + Cz + D \rightarrow D = 0$$

$$\varphi_A = -y'(z=0) = -C = -\frac{F\ell^2}{16EI} \text{ ORARIO}$$

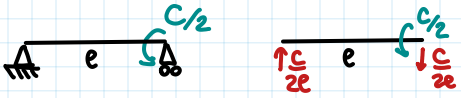
$$y_B = y(z=\frac{\ell}{2}) = -\frac{F}{12}\frac{\ell^3}{8EI} + \frac{F\ell^2}{16EI} \cdot \frac{\ell}{2} = -\frac{F\ell^3}{96EI} + \frac{F\ell^3}{32EI} = \frac{-1+3}{96EI}F\ell^3 = \frac{F\ell^3}{48EI} \oplus \downarrow$$

④ TEMA d'ESAME Giugno 2023

Det. ABBASSAMENTO e ROTAZIONE IN MEZZERIA



R = 1.5 mm
E = 1 GPa

STRUTTURA sym con carico ANTISIMM

$y(B) = 0$